Worksheet 1

Presenting and organising data – EXERCISES

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Exercise 1 – Statistical vocabulary and type of variable

A store surveys 210 customers about the maximum price they are willing to pay for a new product.

1. Identify the observed population, the statistical unit, and the variable studied.
2. State whether the variable is qualitative or quantitative; if quantitative, state whether it is discrete or continuous.
3. After purchase, each customer is asked whether they are *satisfied* or *not satisfied*. What type of variable is this? Which graph is suitable? Are a mean and a standard deviation appropriate?

Exercise 2 – Continuous distribution, histogram, and cumulative counts

The distribution of stated prices is:

Price class (euros)	$[10, 14[$	$[14, 18[$	$[18, 26[$	$[26, 30[$	$[30, 35[$	$[35, 40[$	$[40, 45[$	$[45, 50[$	$[50, 55[$
Count n_i	4	12	32	32	60	35	25	5	5

1. Check the total count. For each class, calculate the class width a_i, relative frequency f_i, and count density $d_i = n_i/a_i$.
2. Draw the correct histogram. Explain why heights must be densities when class widths are unequal.
3. Calculate increasing cumulative counts (ICC) and decreasing cumulative counts (DCC), then the corresponding cumulative relative frequencies.
4. Draw the two cumulative curves.
5. Find the number and proportion of customers willing to pay less than 35 euros, and the number and proportion willing to pay at least 35 euros.

Exercise 3 – Discrete distribution

The number of orders placed in one month by 40 customers is:

x_i	0	1	2	3	4	5
n_i	3	7	12	10	6	2

1. Calculate relative frequencies and increasing cumulative relative frequencies.
2. Draw a bar/stick diagram.
3. Draw the cumulative distribution function and find the proportion of customers who placed at most 3 orders.